# Customer Churn Risk Report

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## Prediction

## **Likely to churn** - churn probability **89.3%**

| | |
|---|---|
| Risk level | High |
| Decision threshold | 55.0% (above this = likely to churn) |
| Retention probability | 10.7% |

The model estimates this customer is at high risk of leaving. Contact them before renewal.

## Customer details

| | |
|---|---|
| Contract type | Month-to-month |
| Payment method | Electronic check |
| Internet service | Fiber optic |
| Online security | No |
| Tech support | No |
| Tenure | 2 months |
| Monthly charges | $95.00 |
| Total charges | $190.00 |

## Key factors

| Input | Value | Effect on estimate |
|---|---|---|
| Contract type | Month-to-month | Pushes the estimate up |
| Tenure | 2 months | Pushes the estimate up |
| Monthly charges | $95.00 | Pushes the estimate up |
| Online security | No | Pushes the estimate up |
| Tech support | No | Pushes the estimate up |

Factors describe patterns the model associates with churn, not proven causes.

## Recommended retention actions

• Offer a discount for moving to a one- or two-year contract.

• Schedule an early-life check-in; newer customers leave more often in this data.

• Review the monthly price against the customer's usage and offer a better-fitting plan.

• Offer a trial of the online-security add-on.

## Model information

| | |
|---|---|
| Model | Gradient Boosting |
| Evaluation | 1409 held-out customers (stratified 20% test split, seed 42) |
| Accuracy / precision / recall | 76.2% / 53.7% / 75.4% |
| F1-score | 62.7% |
| ROC-AUC | 0.838 |

| | |
|---|---|
| PR-AUC (baseline = churn rate) | 0.657 (baseline 0.265) |
| Brier score (lower is better) | 0.164 (baseline 0.195) |
| Confusion matrix (TN/FP/FN/TP) | 792 / 243 / 92 / 282 |
| Data | Telco-Customer-Churn.csv (7,043 customers) |

## How to read the probability

• The predicted outcome compares the estimated probability with the decision threshold.

• Risk bands: Low below 40%, Moderate from 40% to below 70%, High from 70%.

• Key factors list the inputs that moved this customer's estimate the most.

## Limitations

• The model was trained on one public telecom dataset (IBM Telco Customer Churn). Results may not transfer to other businesses, regions or time periods.

• The model does not see support history, usage, contract dates or competitor offers, which often matter for churn.

• Key factors show patterns the model associates with churn. They are not proven causes.

• Probabilities are estimates. Compare customers with each other rather than reading any single score as an exact frequency.

*This report is generated by a statistical model trained on a public telecom dataset. The churn probability is an estimate of how similar customers behaved in that data. It is not a guarantee of what this customer will do, and it does not prove why they would leave. Use it together with your own knowledge of the customer.*